

CHAPTER 6

Signal Generators

LEARNING OBJECTIVES

Upon completion of this chapter on signal generators, you will be able to:

- Explain the operation of a multivibrator circuit, sketch its output voltage waveshapes, and calculate its frequency of oscillation.
- Make a one-shot multivibrator and explain the purpose of this circuit.
- Show how two op amps, three resistors, and one capacitor can be connected to form an inexpensive triangle/square-wave generator.
- Predict the frequency of oscillation and amplitude of the voltages in a bipolar or unipolar triangle-wave generator and identify its disadvantages.
- Build a sawtooth wave generator and tell how it can be used as a voltage-to-frequency converter, frequency modulator, or frequency shift key circuit.
- Connect an AD630 balanced modulator/demodulator to operate as a switched gain amplifier.

- Connect the AD630 to an op amp circuit to make a *precision* triangle-wave generator whose output voltage amplitude can be adjusted independently of the oscillating frequency, and vice versa.
- Build, test, measure, and explain the operation of an AD639 universal trigonometric function generator when it is wired to generate sine functions.
- Connect the AD639 to the triangle-wave generator to make a superb *precision* sine-wave generator. Its oscillating frequency can be adjusted over a wide frequency range by a *single* resistor, without changing amplitude.
- Know about the operation of a single IC function generator.

6-0 INTRODUCTION

Up to now our main concern has been to use the op amp in circuits that process signals. In this chapter we concentrate on op amp circuits that *generate* signals. Four of the most common and useful signals are described by their shape when viewed on an oscilloscope. They are the square wave, triangular wave, sawtooth wave, and sine wave. Accordingly, the signal generator is classified by the shape of the wave it generates. Some circuits are so widely used that they have been assigned a special name. For example, the first circuit presented in Section 6-1 is a multivibrator that generates primarily square waves and exponential waves. Some ICs that generate these waveforms from a single IC are available. However, you may need a waveform quickly and not have on hand one of these function generator ICs.

6-1 FREE-RUNNING MULTIVIBRATOR

6-1.1 Multivibrator Action

A *free-running* or *astable multivibrator* is a square-wave generator. The circuit of Fig. 6-1 is a multivibrator circuit and looks something like a comparator with hysteresis (Chapter 4), except that the input voltage is replaced by a capacitor. Resistors R_1 and R_2 form a voltage divider to feed back a fraction of the output to the (+) input. When V_o is at $+V_{sat}$, as shown in Fig. 6-1(a), the feedback voltage is called the upper-threshold voltage V_{UT} . V_{UT} is given in Eq. (4-1) and repeated here for convenience:

$$V_{UT} = \frac{R_2}{R_1 + R_2} (+V_{sat}) \quad (6-1)$$

Resistor R_f provides a feedback path to the (−) input. When V_o is at $+V_{sat}$, current I^+ flows through R_f to charge capacitor C toward V_{UT} . As long as the capacitor voltage V_C is less than V_{UT} , the output voltage remains at $+V_{sat}$.

When V_C charges to a value slightly greater than V_{UT} , the (−) input goes positive with respect to the (+) input. This switches the output from $+V_{sat}$ to $-V_{sat}$. The (+) input is now held negative with respect to ground because the feedback voltage is negative and given by

$$V_{LT} = \frac{R_2}{R_1 + R_2}(-V_{\text{sat}}) \quad (6-2)$$

Equation (6-2) is the same as Eq. (4-2). Just after V_o switches to $-V_{\text{sat}}$, the capacitor has an initial voltage equal to V_{UT} [see Fig. 6-1(b)]. Now current I^- discharges C to 0 V and recharges C to V_{LT} . When V_C becomes slightly more negative than the feedback voltage V_{LT} , output voltage V_o switches back to $+V_{\text{sat}}$. The condition in Fig. 6-1(a) is reestab-

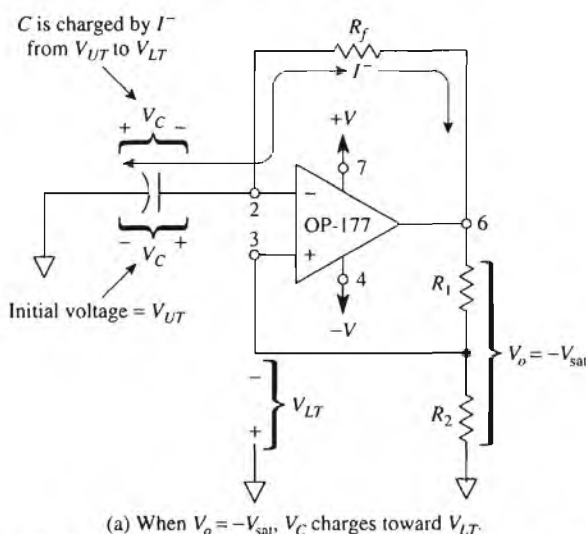
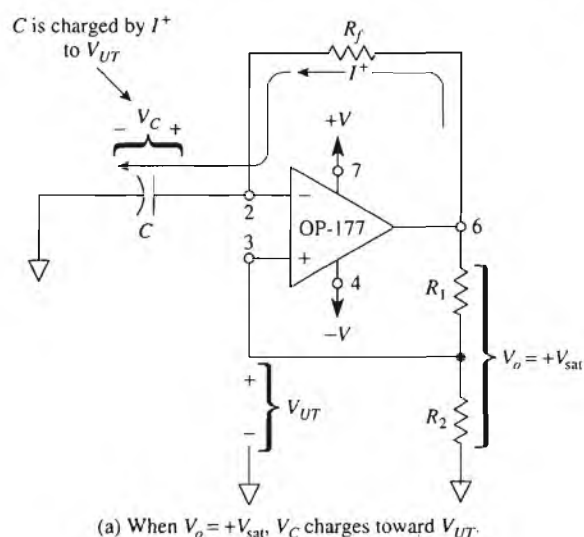


FIGURE 6-1 Free-running multivibrator ($R_1 = 100 \text{ k}\Omega$, $R_2 = 86 \text{ k}\Omega$). Output-voltage waveforms shown in Fig. 6-2.

lished except that C now has an initial charge equal to V_{LT} . The capacitor will discharge from V_{LT} to 0 V and then recharge to V_{UT} , and the process is repeating. Free-running multivibrator action is summarized as follows:

1. When $V_o = -V_{sat}$, C discharges from V_{UT} to V_{LT} and switches V_o to $+V_{sat}$.
2. When $V_o = +V_{sat}$, C charges from V_{LT} to V_{UT} and switches V_o to $-V_{sat}$.

The time needed for C to charge and discharge determines the frequency of the multivibrator.

6-1.2 Frequency of Oscillation

The capacitor and output-voltage waveforms for the free-running multivibrator are shown in Fig. 6-2. Resistor R_2 is chosen to equal $0.86R_1$ to simplify calculation of capacitor charge time. Time intervals t_1 and t_2 show how V_C and V_o change with time for Figs. 6-1(a) and (b), respectively. Time intervals t_1 and t_2 are equal to the product of R_f and C .

The period of oscillation, T , is the time needed for one complete cycle. Since T is the sum of t_1 and t_2 ,

$$T = 2R_f C \quad \text{for } R_2 = 0.86R_1 \quad (6-3a)$$

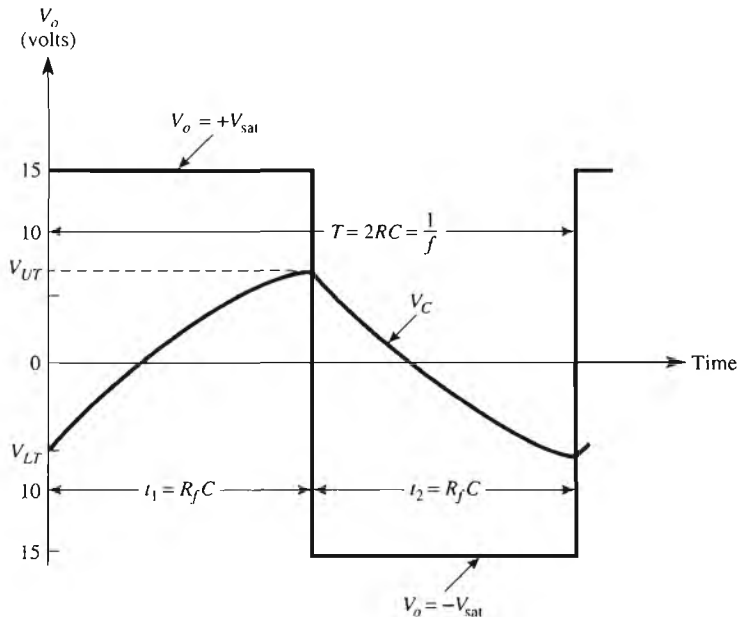


FIGURE 6-2 Voltage waveshapes for the multivibrator of Fig. 6-1.

The frequency of oscillation f is the reciprocal of period T and is expressed by

$$f = \frac{1}{T} = \frac{1}{2R_f C} \quad (6-3b)$$

where T is in seconds, f in hertz, R_f in ohms, and C in farads.

Example 6-1

In Fig. 6-1, if $R_1 = 100 \text{ k}\Omega$, $R_2 = 86 \text{ k}\Omega$, $+V_{\text{sat}} = +15 \text{ V}$, and $-V_{\text{sat}} = -15 \text{ V}$, find (a) V_{UT} ; (b) V_{LT} .

Solution (a) By Eq. (6-1),

$$V_{UT} = \frac{86 \text{ k}\Omega}{186 \text{ k}\Omega} \times 15 \text{ V} \cong 7 \text{ V}$$

(b) By Eq. (6-2),

$$V_{LT} = \frac{86 \text{ k}\Omega}{186 \text{ k}\Omega}(-15 \text{ V}) \cong -7 \text{ V}$$

Example 6-2

Find the period of the multivibrator in Example 6-1 if $R_f = 100 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$.

Solution Using Eq. (6-3a), $T = (2)(100 \text{ k}\Omega)(0.1 \text{ }\mu\text{F}) = 0.020 \text{ s} = 20 \text{ ms}$.

Example 6-3

Find the frequency of oscillation for the multivibrator of Example 6-2.

Solution From Eq. (6-3b),

$$f = \frac{1}{20 \times 10^{-3} \text{ s}} = 50 \text{ Hz}$$

Example 6-4

Show why $T = 2 R_f C$ when $R_2 = 0.86R$, as stated in Eq. (6-3a).

Solution The time required for a capacitor C to charge through a resistor R_f from some *starting* capacitor voltage toward some *aiming* voltage to a *stop* voltage is expressed generally as

$$t = R_f C \ln \left(\frac{\text{aim} - \text{start}}{\text{aim} - \text{stop}} \right)$$

Applying the equation to Fig. 6-2 yields

$$t_1 = R_f C \ln \left(\frac{+V_{\text{sat}} - V_{LT}}{+V_{\text{sat}} - V_{UT}} \right)$$

If the magnitudes of $+V_{\text{sat}}$ and $-V_{\text{sat}}$ are equal, the term in parentheses simplifies to

$$\ln \left[\frac{+V_{\text{sat}} - \frac{R_2}{R_1 + R_2}(-V_{\text{sat}})}{+V_{\text{sat}} - \frac{R_2}{R_1 + R_2}(+V_{\text{sat}})} \right] = \ln \left(\frac{R_1 + 2R_2}{R_1} \right)$$

Since $\ln 2.718 = 1$, the $\ln$ term can be reduced to 1 if

$$\frac{R_1 + 2R_2}{R_1} = 2.718 \quad \text{or} \quad R_2 = 0.86R_1$$

Now $t_1 = R_f C$ and $t_2 = R_f C$ if $R_2 = 0.86R_1$. Therefore, $T = t_1 + t_2 = 2R_f C$.

6-2 ONE-SHOT MULTIVIBRATOR

6-2.1 Introduction

A *one-shot multivibrator* generates a single output pulse in response to an input signal. The length of the output pulse depends only on external components (resistors and capacitors) connected to the op amp. As shown in Fig. 6-3, the one-shot generates a single output pulse on the negative-going edge of E_i . The duration of the input pulse can be longer or shorter than the expected output pulse. The duration of the output pulse is represented by τ in Fig. 6-3. Since τ can be changed only by changing resistors or capacitors, the one-shot can be considered a *pulse stretcher*. This is because the width of the pulse can be longer than the input pulse. Moreover, the one-shot introduces an idea of an adjustable delay, that is, the delay between the time when E_i goes negative and the time for V_o to go positive again. Operation of the one-shot will be studied in three parts: (1) the stable state, (2) transition to the timing state, and (3) the timing state.

6-2.2 Stable State

In Fig. 6-4(a), V_o is at $+V_{\text{sat}}$. Voltage divider R_1 and R_2 feeds back V_{UT} to the (+) input. V_{UT} is given by Eq. (6-1). The diode D_1 clamps the (−) input at approximately $+0.5$ V. The (+) input is positive with respect to the (−) input, and the high open-loop gain times the differential input voltage ($E_d = 2.1 - 0.5 = 1.6$ V) holds V_o at $+V_{\text{sat}}$.

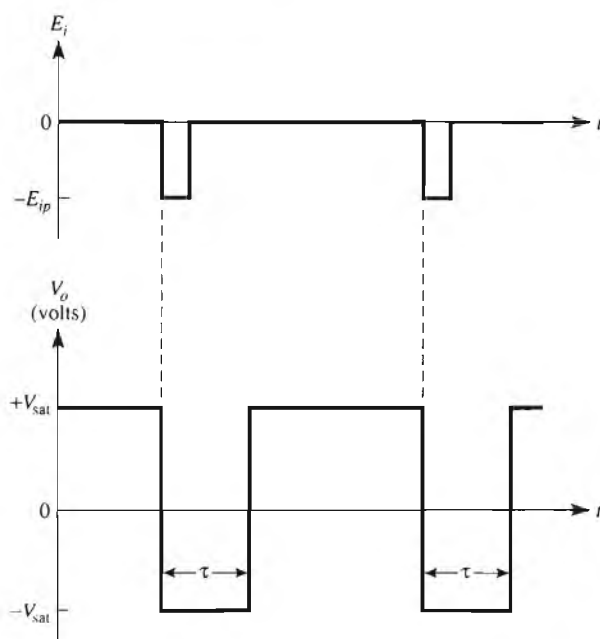


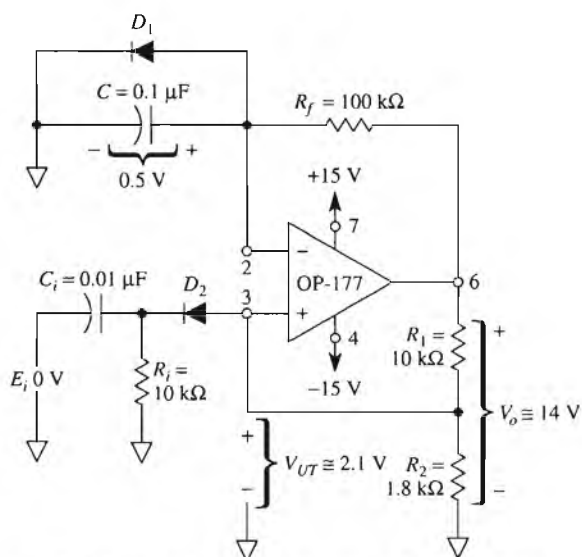
FIGURE 6-3 Input signal E_i and output pulse of a one-shot multivibrator.

6-2.3 Transition to the Timing State

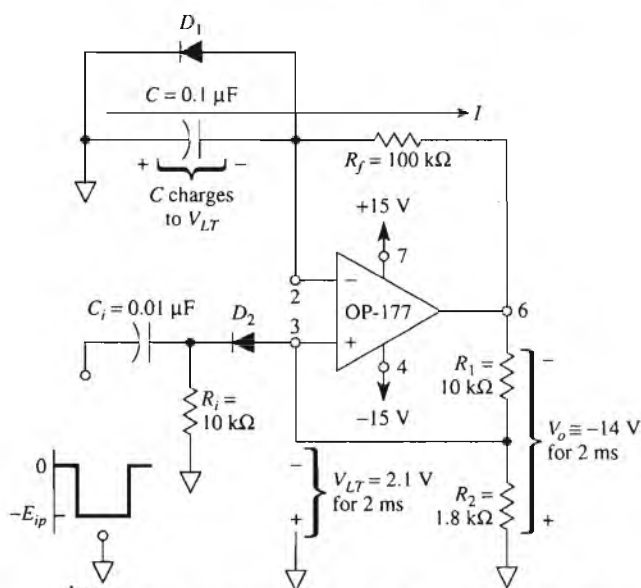
If input signal E_i is at a steady dc potential as in Fig. 6-4(a), the (+) input remains positive with respect to the (-) input and V_o stays at $+V_{sat}$. However, if E_i goes negative by a peak value E_{ip} approximately equal to or greater than V_{UT} , the voltage at the (+) input will be pulled below the voltage at the (-) input. Once the (+) input becomes negative with respect to the (-) input, V_o switches to $-V_{sat}$. With this change, the one-shot is now in its timing state. For best results, the time constant $R_i C_i$ should be 1/10 the time constant $R_f C$ or less.

6-2.4 Timing State

The timing state is an unstable state; that is, the one-shot cannot remain very long in this state for the following reasons. Resistors R_1 and R_2 in Fig. 6-4(b) feed back a negative voltage ($V_{LT} = -2.1$ V) to the (+) input. The diode D_1 is now reverse biased by $-V_{sat}$ and is essentially an open circuit. Capacitor C discharges to 0 and then recharges with a polarity opposite to that in Fig. 6-4(a) [see Fig. 6-4(b)]. As C recharges, the (-) input becomes more and more negative with respect to ground. When the capacitor voltage is slightly more negative than V_{LT} , V_o switches to $+V_{sat}$. The one-shot has now completed its output pulse and is back to the stable state in Fig. 6-4(a). Since the one-shot has only one stable state, it is also called a *monostable multivibrator*.



(a) Stable state of a one-shot multivibrator: $R_i C_i \leq (R_f C)/10$.



(b) Timing state: When E_i goes negative, V_o goes negative for 2 ms.

FIGURE 6-4 Monostable or one-shot multivibrator.

6-2.5 Duration of Output Pulse

If R_2 is made about one-fifth of R_1 , in Fig. 6-4, then the duration of output pulse is given by

$$\tau = \frac{R_f C}{5} \quad \text{if } R_2 = \frac{R_1}{5} \quad (6-4a)$$

The exact equation is

$$\tau = R_f C \ln \left[\frac{-V_{sat} - 0.5 \text{ V}}{-V_{sat} - V_{UT}} \right] \quad (6-4b)$$

Example 6-5

Calculate τ for the one-shot of Fig. 6-4.

Solution By Eq. (6-4a),

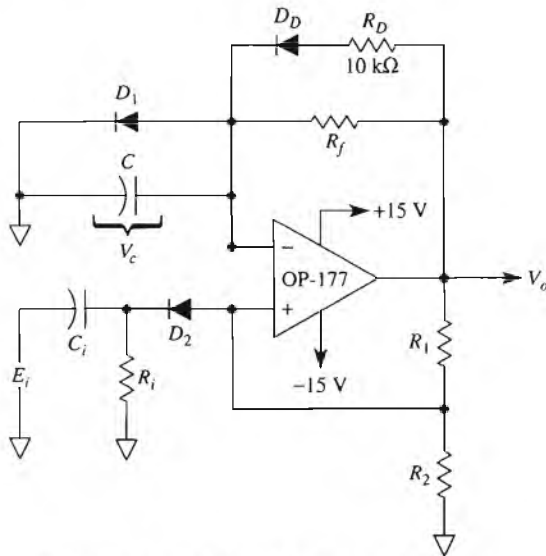
$$\tau = \frac{(100 \text{ k}\Omega)(0.1 \text{ }\mu\text{F})}{5} = 2 \text{ ms}$$

For test purposes, E_i can be obtained from a square-wave or pulse generator. Diode D_2 prevents the one-shot from coming out of the timing state on positive transitions of E_i . To build a one-shot that has a positive output pulse for a positive input signal, simply reverse the diodes.

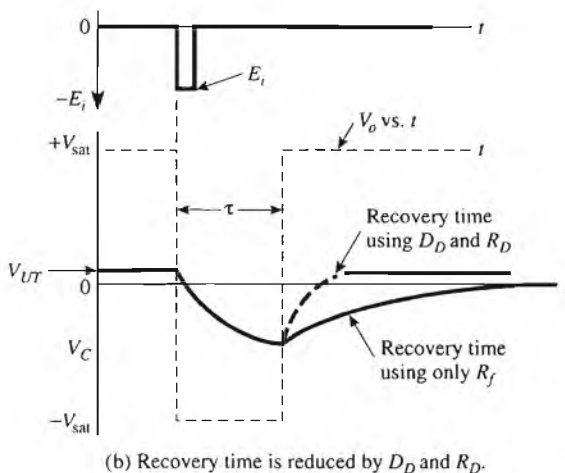
6-2.6 Recovery Time

After the timing state is completed, the output returns to $+V_{sat}$. However, the circuit is not ready to be retriggered reliably until C returns to its initial state of 0.5 V because it takes time for C to be discharged from $V_{LT} = -2.1 \text{ V}$ in Fig. 6-4(b) to 0.5 V in Fig. 6-4(a). This time interval is called *recovery time* and is shown in Fig. 6-5(b). Recovery time is approximately τ .

Normally, C is charged back to its initial state by a current through R_f . By adding a discharge resistor R_D in parallel with R_f , as in Fig. 6-5(a), the recovery time is reduced. Typically, if $R_D = 0.1R_f$, recovery time is reduced to one-tenth. Diode D_D prevents R_D from affecting the timing-cycle interval τ .



(a) Discharge diode D_D and R_D are added to Fig. 6-4.



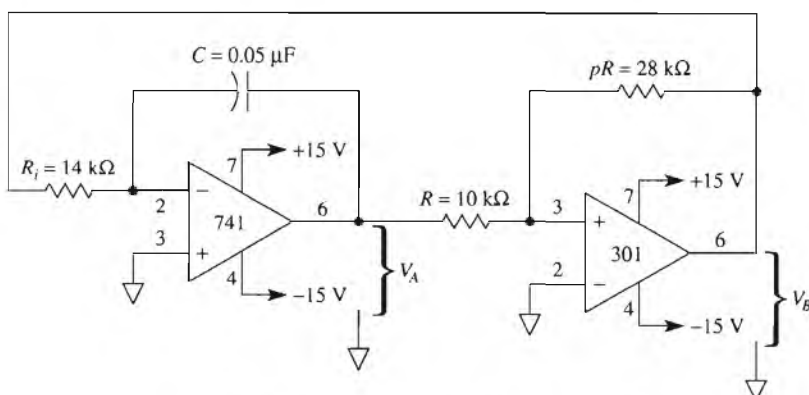
(b) Recovery time is reduced by D_D and R_D .

FIGURE 6-5 The recovery time of a one-shot multivibrator is reduced by adding discharge diode D_D and R_D . R_D should be about one-tenth of R_f to reduce recovery time to one-tenth.

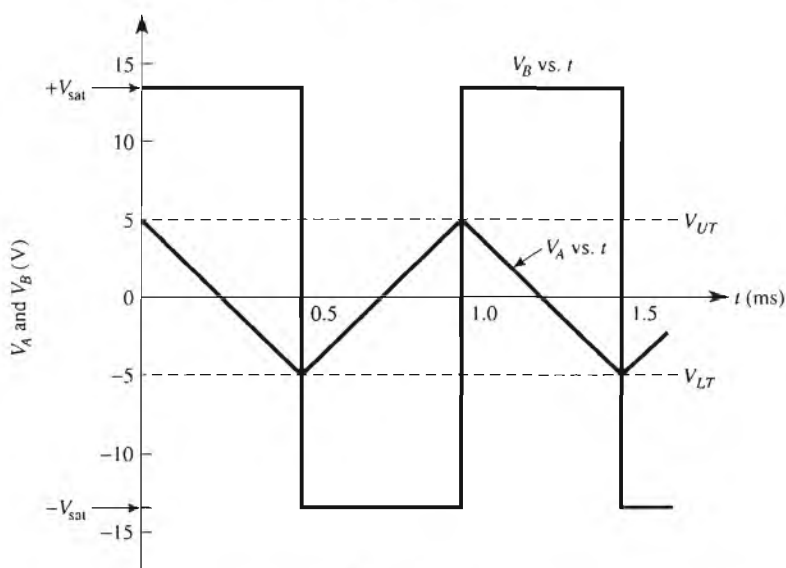
6-3 TRIANGLE-WAVE GENERATORS

6-3.1 Theory of Operation

A basic bipolar triangle-wave generator circuit is presented in Fig. 6-6. The triangle wave, V_A , is available at the output of the 741 integrator circuit. An additional square-wave signal, V_B , is available at the output of the 301 comparator.



(a) The 741 integrator circuit and 301 comparator circuit are wired to make a triangle-wave generator.



(b) Waveshapes.

FIGURE 6-6 The bipolar triangle-wave generator circuit in (a) generates triangle-wave and square-wave oscillator signals as in (b). (a) Basic bipolar triangle-wave generator oscillator frequency for 1000 Hz; (b) output-voltage waveshapes.

To understand circuit operation, refer to time interval 0 to 1 ms in Fig. 6-6. Assume that V_B is high at $+V_{sat}$. This forces a constant current (V_{sat}/R_i) through C (left to right) to drive V_A negative from V_{UT} to V_{LT} . When V_A reaches V_{LT} , pin 3 of the 301 goes negative, and V_B snaps to $-V_{sat}$, and $t = 1$ ms.

When V_B is at $-V_{\text{sat}}$, it forces a constant current (right to left) through C to drive V_A positive from V_{LT} toward V_{UT} (see the time interval 1 to 2 ms). When V_A reaches V_{UT} at $t = 2$ ms, pin 3 of the 301 goes positive and V_B snaps to $+V_{\text{sat}}$. This initiates the next cycle of oscillation.

6-3.2 Frequency of Operation

The peak values of the triangular wave are established by the *ratio* of resistor pR to R and the saturation voltages. They are given by

$$V_{UT} = -\frac{-V_{\text{sat}}}{p} \quad (6-5a)$$

$$V_{LT} = -\frac{+V_{\text{sat}}}{p} \quad (6-5b)$$

where

$$p = \frac{pR}{R} \quad (6-5c)$$

If the saturation voltages are reasonably equal, the frequency of oscillation, f , is given by

$$f = \frac{p}{4R_i C} \quad (6-6)$$

Example 6-6

A triangle-wave generator oscillates at a frequency of 1000 Hz with peak values of approximately ± 5 V. Calculate the required values for pR , R_i , and C in Fig. 6-6.

Solution First we work on the calculation for the comparator resistor ratio p that controls peak triangle-wave output voltages, V_{UT} and V_{LT} . $+V_{\text{sat}}$ is practically $+14.2$ V and $-V_{\text{sat}}$ is typically -13.8 V for a ± 15 -V supply. This observation points out one deficiency in our low-cost triangle-wave generator. It does *not* have *precisely* equal positive and negative peak outputs. (We will remedy this problem, at a higher cost, in Section 6-6.) From Eq. (6-5a), solve for p :

$$p = -\frac{-V_{\text{sat}}}{V_{UT}} = -\frac{-13.8 \text{ V}}{5 \text{ V}} = +2.76 \approx 2.8$$

Choose $R = 10 \text{ k}\Omega$. Then from Eq. (6-5c) we solve for pR as

$$pR = 2.8(10 \text{ k}\Omega) = 28 \text{ k}\Omega$$

Next we select R_i and C . Begin by making a *trial* choice for $C = 0.05 \mu\text{F}$. Then calculate a value for R_i to see if R_i is greater than $10 \text{ k}\Omega$. From Eq. (6-6),

$$R_i = \frac{p}{4fC} = \frac{2.8}{4(1000 \text{ Hz})(0.05 \mu\text{F})} = 14 \text{ k}\Omega$$

In practice it would be prudent for R_i to be a $12\text{-k}\Omega$ resistor in series with a 0 to $5\text{-k}\Omega$ pot. The $5\text{-k}\Omega$ pot may then be adjusted for an oscillation frequency of precisely 1.00 kHz .

6-3.3 Unipolar Triangle-Wave Generator

The bipolar triangle-wave generator circuit of Fig. 6-6 can be changed to produce a unipolar triangle wave output. Simply add a diode in series with pR as shown in Fig. 6-7. Circuit operation is studied by reference to the waveshapes in Fig. 6-7(b).

When V_B is at $+V_{\text{sat}}$, the diode stops current flow through pR and sets V_{LT} at 0 V . When V_B is at $-V_{\text{sat}}$, the diode allows current flow through pR and sets V_{UT} at a value of

$$V_{UT} = -\frac{-V_{\text{sat}} + 0.6 \text{ V}}{p} \quad (6-7a)$$

Frequency of oscillation is then given approximately by

$$f \approx \frac{p}{2R_i C} \quad (6-7b)$$

Example 6-7

Find the approximate peak voltage and frequency for the unipolar triangle-wave generator in Fig. 6-7.

Solution Calculate

$$p = \frac{pR}{R} = \frac{28 \text{ k}\Omega}{10 \text{ k}\Omega} = 2.8$$

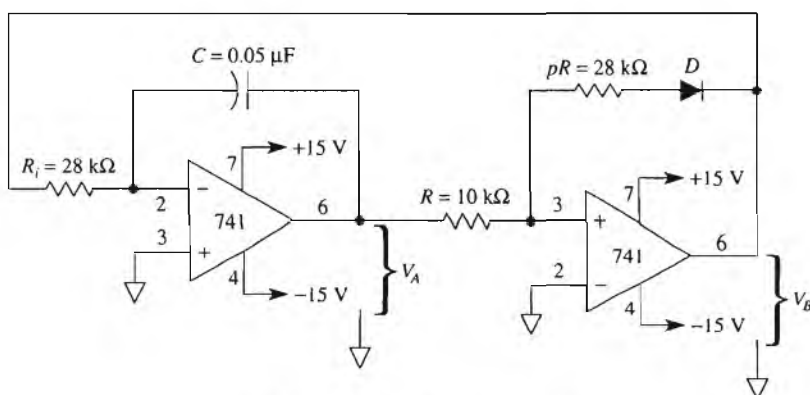
Find the peak value of V_A from Eq. (6-7a):

$$V_{UT} = -\left(\frac{-V_{\text{sat}} + 0.6 \text{ V}}{p}\right) = -\left(\frac{-13.8 \text{ V} + 0.6 \text{ V}}{2.8}\right) \approx 4.7 \text{ V}$$

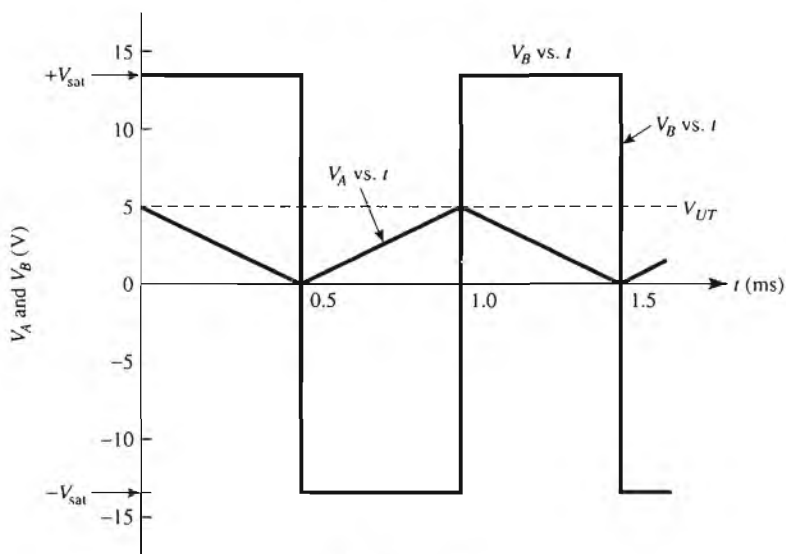
From Eq. (6-7b),

$$f = \frac{p}{2R_i C} = \frac{2.8}{2(28 \text{ k}\Omega)(0.05 \mu\text{F})} = 1000 \text{ Hz}$$

Note the change in R_i value from Fig. 6-6(a) to Fig. 6-7(a).



(a) Unipolar triangle-wave generator.



(b) Waveshapes.

FIGURE 6-7 Diode D in (a) converts the bipolar triangle-wave generator into a unipolar triangle-wave generator. Waveshapes are shown in (b). (a) Basic unipolar triangle-wave generator; oscillating frequency is 1000 Hz . (b) Output-voltage waveshapes.

6-4 SAWTOOTH-WAVE GENERATOR

6-4.1 Circuit Operation

A low-parts-count sawtooth-wave generator circuit is shown in Fig. 6-8(a). Op amp *A* is a ramp generator. Since E_i is negative, V_{oramp} can only ramp up. The rate of rise of the ramp voltage is constant at

$$\frac{V_{oramp}}{t} = \frac{E_i}{R_i C} \quad (6-8)$$

The ramp voltage is monitored by the (+) input of comparator 301B. If V_{oramp} is below V_{ref} , the comparator's output is negative. Diodes protect the transistors against excessive reverse bias.

When V_{oramp} rises to just exceed V_{ref} , the output V_{ocomp} goes to positive saturation. This forward biases "dump" transistor Q_D into saturation. The saturated transistor acts as a short circuit across the integrating capacitor C . C discharges quickly through Q_D to essentially 0 V. When V_{ocomp} goes positive, it turns on Q_1 to short-circuit the 10-k Ω potentiometer. This drops V_{ref} to almost zero volts.

As C discharges toward 0 V, it drives V_{oramp} rapidly toward 0 V. V_{oramp} drops below V_{ref} , causing V_{ocomp} to go negative and turn off Q_D . C begins charging linearly, and generation of a new sawtooth wave begins.

6-4.2 Sawtooth Waveshape Analysis

The ramp voltage rises at a rate of 1 V per millisecond in Fig. 6-8(b). Meanwhile, V_{ocomp} is shown to be negative. When the ramp crosses V_{ref} , V_{ocomp} snaps positive to drive the ramp voltage quickly toward 0 V. As V_{oramp} snaps to 0 V, the comparator's output is reset to negative saturation. Ramp operation is summarized in Fig. 6-8(c).

6-4.3 Design Procedure

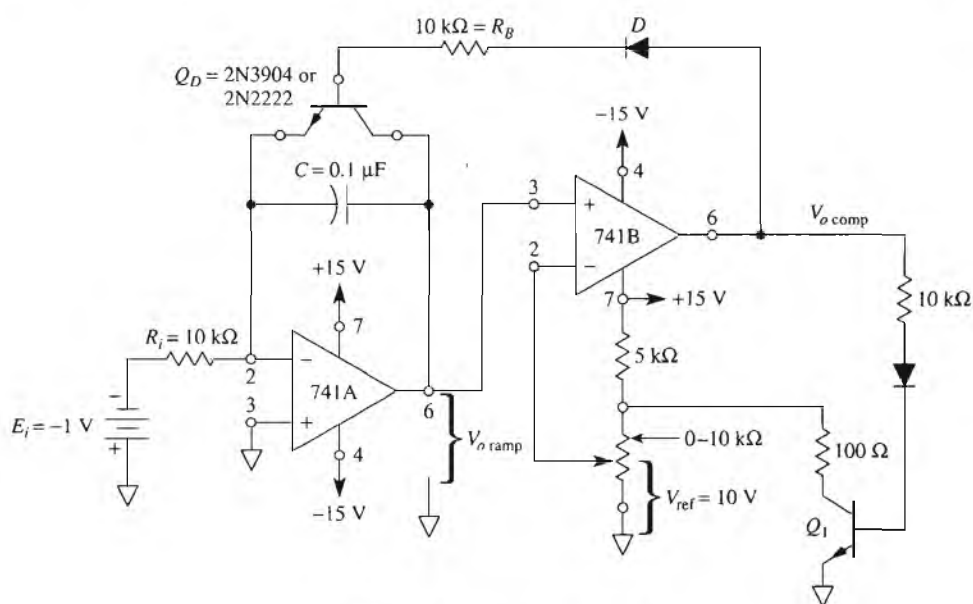
The time for one sawtooth-wave period can be derived most efficiently by analogy with a familiar experience.

$$\text{time (of rise)} = \frac{\text{distance (of rise)}}{\text{speed (of rise)}} \quad (6-9a)$$

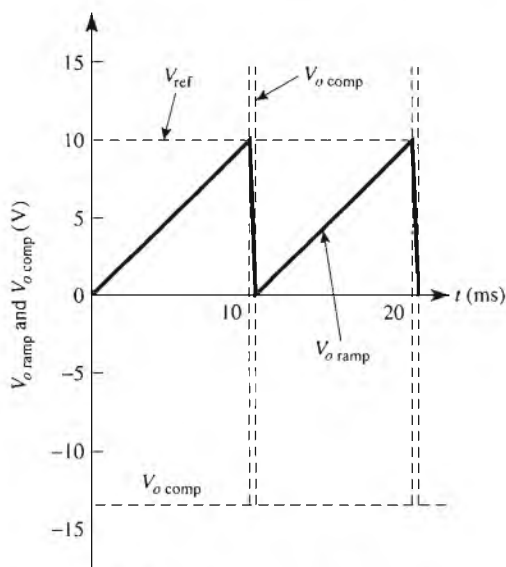
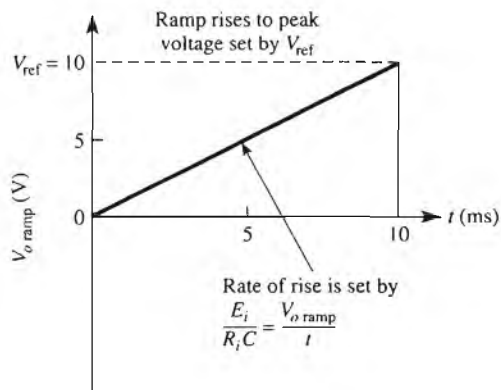
$$\text{period } T = \frac{V_{ref}}{E_i/R_i C} \quad (6-9b)$$

Since frequency is the reciprocal of the period

$$f = \left(\frac{1}{R_i C} \right) \frac{E_i}{V_{ref}} \quad (6-9c)$$



(a) Sawtooth-wave generator circuit.

(b) Sawtooth-wave output $V_{o \text{ ramp}}$ and comparator output.

(c) Details for sawtooth-wave design or analysis.

FIGURE 6-8 The sawtooth-wave generator circuit in (a) has the waveshapes shown in (b) and (c). Oscillating frequency is 100 Hz or $f = (1/R_i C)(E_i/V_{ref})$.

Design Example 6-8

Design a sawtooth-wave generator to have a 10-V peak output and a frequency of 100 Hz. Let $E_i = 1$ V.

Design Procedure

1. Design a voltage divider to give a reference voltage $V_{\text{ref}} = +10$ V for op amp B in Fig. 6-8(a).
2. Let's select a ramp rate rise of 1 V/ms. Pick any $R_i C$ combination to give 1.0 ms. Therefore, let's select $R_i = 10$ k Ω and $C = 0.1$ μ F.
3. The resulting circuit is shown in Fig. 6-8(a).
4. E_i may be made from a voltage divider and voltage follower to make an ideal voltage source (see Section 3.7).
5. Alternatively, you could pick a trial value for $R_i C$ and solve for E_i in Eq. (6-9b).
6. Check the design values in Eq. 6-9c.

$$f = \frac{1}{(10 \text{ k}\Omega)(0.1 \text{ }\mu\text{F})} \left(\frac{1 \text{ V}}{10 \text{ V}} \right) = 100 \text{ Hz}$$

6-4.4 Voltage-to-Frequency Converter

There are two ways to change or modulate the oscillating frequency of Fig. 6-8. We see from Eq. (6-9c) that the frequency is directly proportional to E_i and inversely proportional to V_{ref} . The advantages and disadvantages of each method are examined with an example.

This type of frequency modulation by V_{ref} has two disadvantages with respect to control of frequency by E_i . First, the relationship between input voltage V_{ref} and output frequency is *not* linear. Second, the sawtooth's peak output voltage is not constant, since it varies directly with V_{ref} .

6-4.5 Frequency Modulation and Frequency Shift Keying

Examples 6-9 and 6-10 indicate one way of achieving *frequency modulation* (FM). Thus, if the amplitude of E_i varies, the frequency of the sawtooth oscillator will be changed or modulated. If E_i is keyed between two voltage levels, the sawtooth oscillator changes frequencies. This type of application is called *frequency shift keying* (FSK) and is used for data transmission. These two preset frequencies correspond to "0" and "1" states (commonly called *space* and *mark*) in binary.

Example 6-9

If E_i is doubled to -2 V in Fig. 6-8, find the new frequency of oscillation.

Solution In Eq. (6-9c) use $|E_i|$

$$f = \left(\frac{1}{R_i C} \right) \frac{E_i}{V_{\text{ref}}} = \frac{1}{(10 \times 10^3 \Omega)(0.1 \times 10^{-6} \text{ F})} \frac{E_i}{10 \text{ V}}$$

$$= \frac{1}{1.0 \times 10^{-3} \text{ s}} = \frac{E_i}{10 \text{ V}} = \left(\frac{100 \text{ Hz}}{\text{V}} \right) E_i$$

For $E_i = -2$ V, $f = (2 \text{ V})(100 \text{ Hz/V}) = 200 \text{ Hz}$. Thus as E_i changes from 0 V to -10 V, frequency changes from 0 Hz to 1 kHz. The peak amplitude of the sawtooth wave remains equal to V_{ref} (10 V) for all frequencies.

Example 6-10

Keep E_i , R_i , and C at their value shown in Fig. 6-8(a). Reduce V_{ref} by one-half to 5 V. Is the frequency doubled or halved?

Solution From Example 6-8 and Eq. (6-9c),

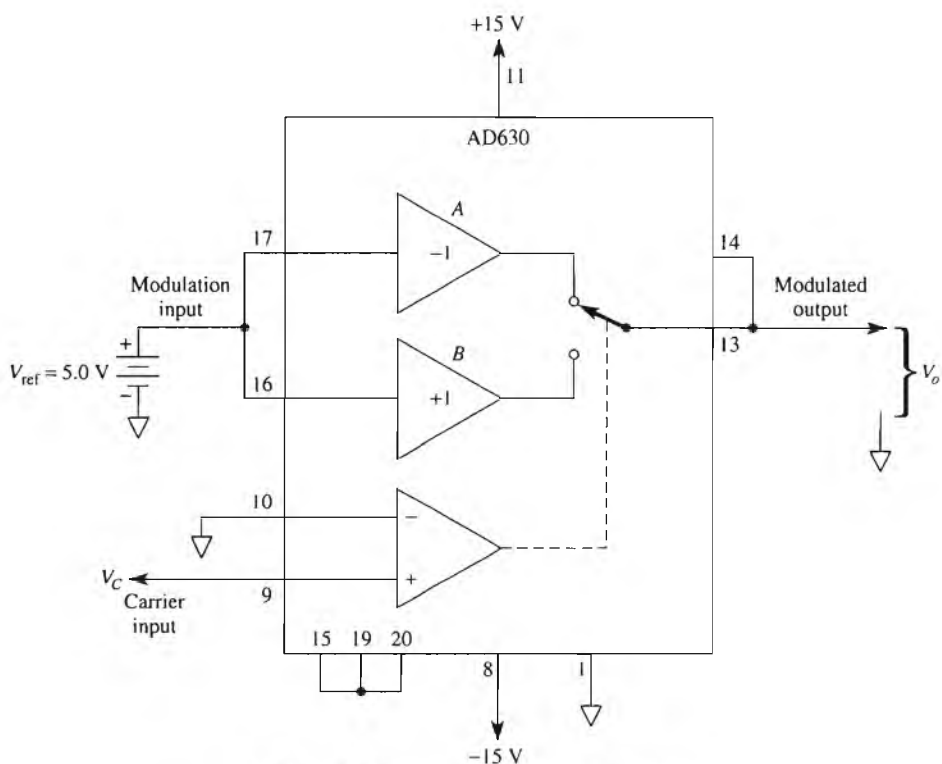
$$f = \frac{1 \text{ V}}{(\text{ms})V_{\text{ref}}} = \frac{(1000 \text{ Hz})/\text{V}}{V_{\text{ref}}}$$

For $V_{\text{ref}} = 10$ V, $f = 100$ Hz. For $V_{\text{ref}} = 5$ V the frequency is doubled to 200 Hz. As V_{ref} is *reduced* from 10 V to 0 V, the frequency is increased from 100 Hz to a very high value.

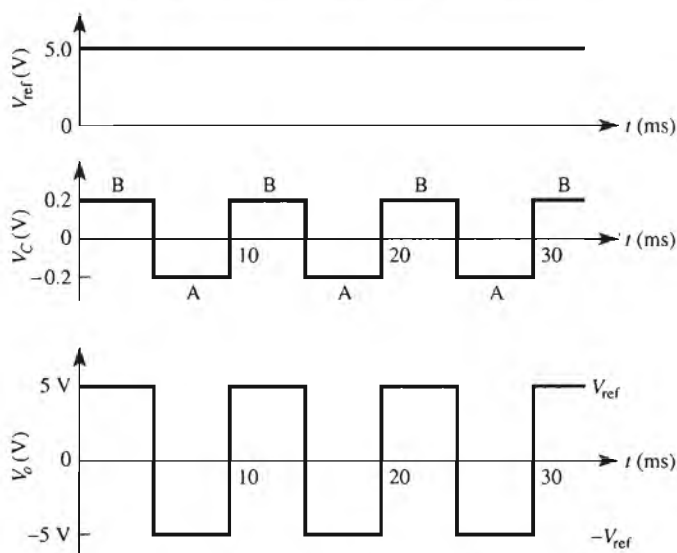
6-4.6 Disadvantages

The triangle-wave generators of Section 6-3 are inexpensive and reliable. However, they have two disadvantages. The rates of rise and fall of the triangle wave are unequal. Also, the peak values of both triangle-wave and square-wave outputs are unequal, because the magnitudes of $+V_{\text{sat}}$ and $-V_{\text{sat}}$ are unequal.

In the next section we substitute an AD630 for the comparator. This will give the equivalent of precisely equal square-wave $\pm$ voltages that will *also* be equal to the $\pm$ peak values of triangle-wave voltage. Once we have made a precision triangle-wave generator, we will use it to drive a new state-of-the-art trigonometric function generator to make a precision sine-wave generator.



(a) The AD630 is wired as a switch gain amplifier.



(b) Output and input waveshapes.

FIGURE 6-9 Operation of the AD630 balanced modulator/demodulator as a switched gain amplifier. (a) Wiring for switched gains of +1 or -1; (b) carrier V_c selects gains of +1 or -1 for input V_{ref} . Output V_o is equal to precisely V_{ref} or $-V_{ref}$.

6-5 BALANCED MODULATOR/DEMODULATOR, THE AD630

6-5.1 Introduction

The AD630 is an advanced integrated circuit. It has 20 pins, which allows this versatile switched voltage gain IC to act as a modulator, demodulator, phase detector, and multiplexer, as well as perform other signal conditioning tasks. We connect the AD630, as in Fig. 6-9(a), as a controlled switched gain (+1 or -1) amplifier. This particular application will be examined by discussing the role performed by the dominant terminals.

6-5.2 Input and Output Terminals

The input signal V_{ref} is connected to *modulation* pins 16 and 17 in Fig. 6-9, and thus to the inputs of two amplifiers, *A* and *B*. The gain of *A* is programmed for -1 and *B* for $+1$ by shorting terminals (1) 13 to 14, (2) 15 to 19 to 20, (3) 16 to 17, and (4) grounding pin 1.

The carrier input terminal, pin 9 (in this application), determines which amplifier, *A* or *B*, is connected to the output terminal. If pin 9 is *above* the voltage at pin 10 (ground), amplifier *B* is selected. Voltage at output pin 13 then equals V_{ref} times $(+1)$.

If pin 9 voltage is *below* ground (negative), amplifier *A* is selected and output pin 13 equals V_{ref} times (-1) . (Note that in communication circuits, V_{ref} is called the *analog data* or *signal voltage*, V_C is called a *chopper* or *carrier voltage*, and V_o is the *modulated* output. That is, the *amplitude* of the low-frequency signal voltage is impressed upon the higher-frequency carrier wave—hence the names selected for the AD630's input and output terminals.)

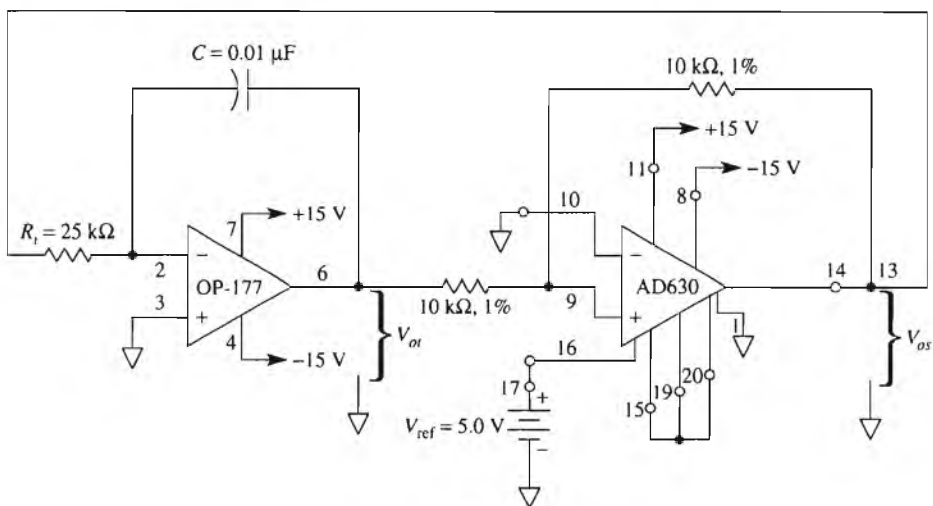
6-5.3 Input–Output Waveforms

V_{ref} is a dc voltage of 5.0 V in Fig. 6-9(b). V_C is a 100-Hz square wave with peak amplitudes that must exceed ± 1 mV. Output voltage V_o is shown in Fig. 6-9(c) to switch synchronously with V_C from $+V_{\text{ref}}$ to $-V_{\text{ref}}$, and vice versa. We are going to replace the unpredictable $\pm V_{\text{sat}}$ of the 301 comparator in Fig. 6-6 with precisely $+V_{\text{ref}}$ or $-V_{\text{ref}}$. Moreover, V_{ref} can be adjusted easily to any required value. As shown in the next section, V_{ref} will set the positive and negative peak values of both triangle-wave and square-wave generators.

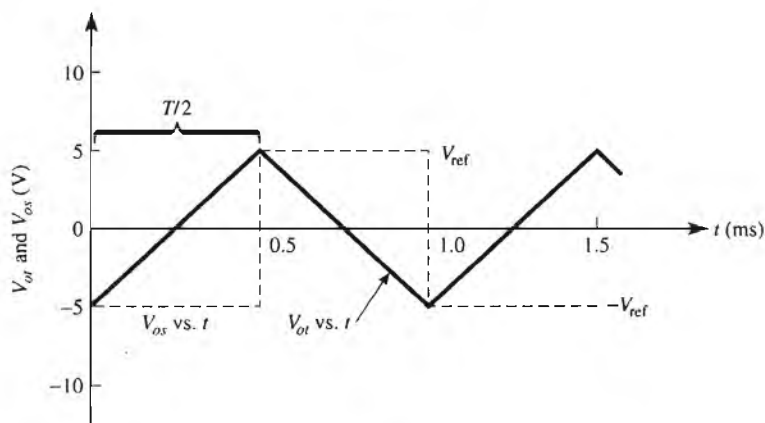
6-6 PRECISION TRIANGLE/SQUARE-WAVE GENERATOR

6-6.1 Circuit Operation

Only six parts plus a voltage source, V_{ref} , make up the versatile precision triangle- and square-wave generator in Fig. 6-10(a). Circuit operation is explained by referencing the waveshapes in Fig. 6-10(b). We begin at time zero. Square-wave output V_{os} begins at $-V_{\text{ref}}$ or -5 V. This forces the triangle wave V_{ot} to go positive from a starting point of $-V_{\text{ref}} = -5$ V. During this time, pin 9 is below ground to select an AD630 gain of -1 and holds V_{os} at -5 V.



(a)



(b)

FIGURE 6-10 The precision triangle/square-wave oscillator in (a) has the output waveshapes in (b). V_{ref} should be buffered for a low-impedance source voltage. (a) Precision triangle/square-wave oscillator (compare with Fig. 6-6). V_{ref} must be a low-impedance source. V_{ref} sets the $\pm$ peak values and R_i adjusts the frequency. (b) Square- and triangular-wave output waveshapes.

At time $T/2 = 0.5$ ms, V_{ot} reaches $+5$ V, where pin 9 is driven slightly positive to select an AD630 gain of $+1$. This snaps V_{os} to $V_{ref} = +5$ V. V_{os} then drives V_{ot} negative. When V_{ot} reaches -5 V, pin 9 goes negative at $T = 1.0$ ms and snaps V_{os} negative to -5 V. This completes one cycle of oscillation and begins another.

6-6.2 Frequency of Oscillation

The easiest way to find the frequency of oscillation is to begin with the rate of rise of the triangle wave, V_{oi}/t , in volts per second. The rate of rise of the triangle wave, from 0 to 0.5 ms in Fig. 6-10(b), is found from

$$\frac{V_{oi}}{t} = \frac{V_{ref}}{R_i C} \quad (6-10)$$

The time t for a half-cycle is $T/2$, and during this time, V_{oi} changes by $2V_{ref}$. Substituting these for t and V_{oi} into Eq. (6-10), we obtain

$$\frac{2V_{ref}}{T/2} = \frac{V_{ref}}{R_i C} \quad (6-11)$$

and solve for both period T and frequency of oscillation f :

$$T = 4R_i C \quad \text{and} \quad f = \frac{1}{T} = \frac{1}{4R_i C} \quad (6-12)$$

Note that V_{ref} cancels out in Eqs. (6-11) and (6-12). This is a very important advantage. The peak output voltages of both square- and triangle-wave signals are set by $+V_{ref}$. As V_{ref} is adjusted, the frequency of oscillation does *not* change.

Example 6-11

Make a triangle/square-wave generator that has peak voltages of ± 5 V and oscillates at a frequency of 1.0 kHz.

Solution Choose $V_{ref} = 5.0$ V. For low impedance, V_{ref} should be the output of an op amp. Arbitrarily choose $C = 0.01$ μ F. From Eq. (6-12),

$$R_i = \frac{1}{4fC} = \frac{1}{4(1000)(0.01 \mu F)} = 25.0 \text{ k}\Omega$$

For a fine adjustment of the output frequency, make R_i from a 22-k Ω resistor in series with a 5- or 10-k Ω variable resistor.

6-7 SINE-WAVE GENERATION SURVEY

Commercial function generators produce triangular, square, and sinusoidal signals whose frequency and amplitude can be changed by the user. To obtain a sine-wave output, the triangle wave is passed through a shaping network made of carefully selected resistors

and diodes (see Fig. 7-19). The sine waves thus produced are reasonably good. However, there is inevitably some distortion, particularly at the peaks of the sine wave. Another solution is to use an IC function generator chip such as MAXIM's MAX038, which is capable of producing sine, square, and triangle waveforms from 1 Hz to 20 MHz.

When an application requires a single-frequency sine wave, conventional oscillators use phase-shifting techniques that usually employ (1) two RC tuning networks, and (2) complex amplitude limiting circuitry. To minimize distortion, the limit circuit must usually be custom-adjusted for each oscillator. The frequency of this oscillator is difficult to vary because two RC networks must be varied and their values must track within $\pm 1\%$.

Waveforms may also be generated by using the AD630 with a universal trigonometric function generator, the AD639. The AD630 has already been used to generate a precision triangle wave whose frequency and amplitudes are precise and easy to adjust. We will connect the triangle-wave output V_{ot} of Fig. 6-10(a) to an AD639 universal trigonometric function generator. The resulting circuit will have the best qualities of a *precision sine-wave generator whose frequency will be easily adjustable*.

6-8 UNIVERSAL TRIGONOMETRIC FUNCTION GENERATOR, THE AD639

6-8.1 Introduction

The AD639 is a state-of-the-art trigonometric function generator. It will perform all trigonometric functions *in real time*, including sin, cos, tan, cosec, sec, and cotan. When a calculator performs a trig function, the operator punches in a number corresponding to the number of angular degrees and punches SIN. The calculator pauses, then displays a number indicating the sine of the angle. That is, a number for angle θ is entered and the calculator produces a number for $\sin \theta$.

The AD639 accepts an input voltage that represents the angle. It is called the *angle voltage*, V_{ang} . For the AD639, the angle voltage is found from

$$V_{ang} = \left(\frac{20 \text{ mV}}{1^\circ} \right) \theta = \left(\frac{1 \text{ V}}{50^\circ} \right) \theta \quad (6-13)$$

Four input terminals are available. However, we shall look at only the single active input that generates sin functions. The output voltage will equal $\sin \theta$ or $10 \sin \theta$, depending how the internal gain control is pin programmed.

6-8.2 Sine Function Operation

The AD639 is wired to output $V_o = 1 \sin \theta$ in Fig. 6-11. There are four input terminals: 1, 2, 7, and 8. Wired as shown, the chip performs a sine function. Pins 3, 4, and 10 control gain. Normally, 3 and 4 are grounded so that pin 10 can activate the internal gain control. A gain of 1 results when pin 10 is wired to $-V_s$ or pin 9. Wire pin 10 to $+V_s$ or pin

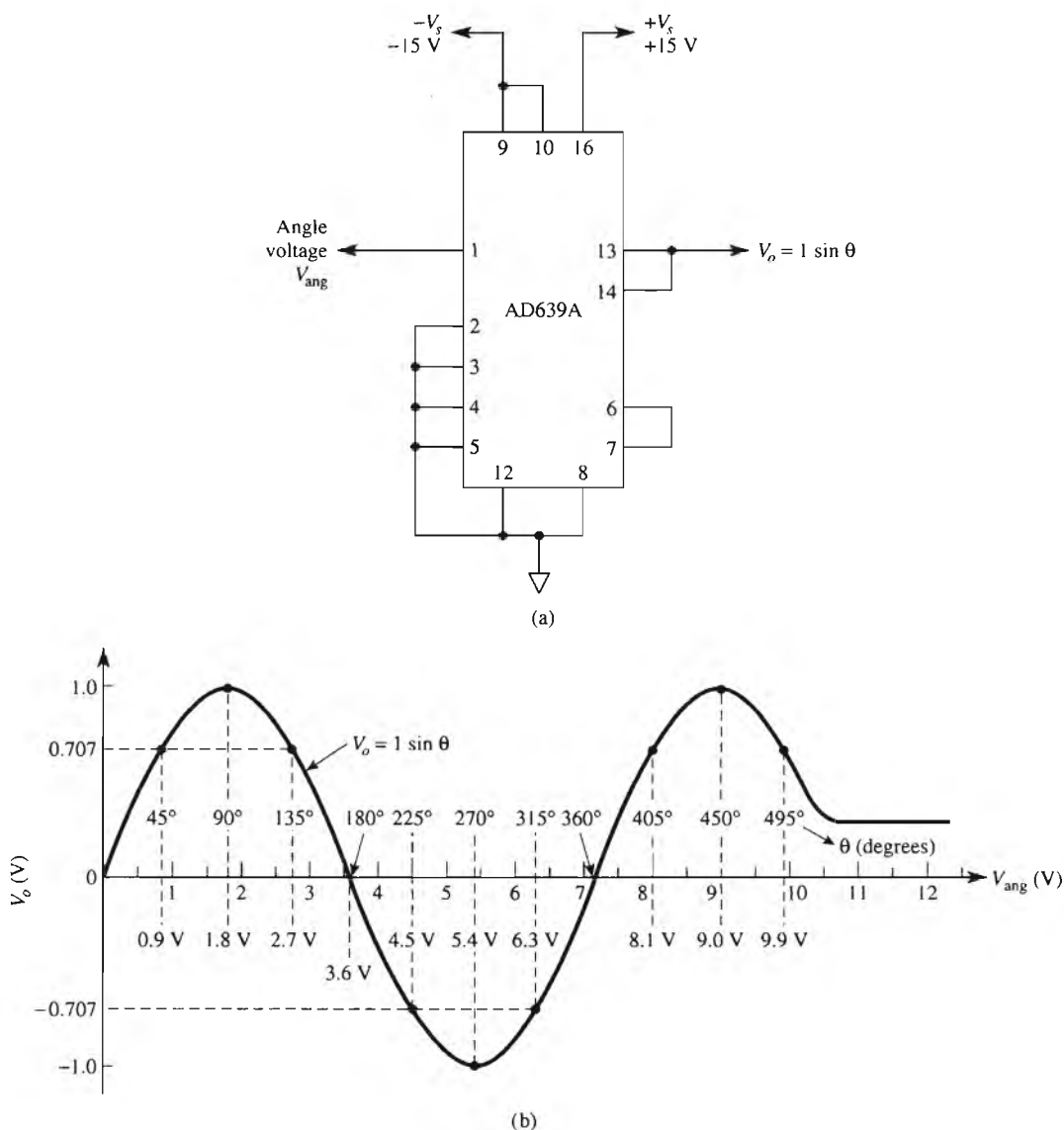


FIGURE 6-11 The AD639 is pin-programmed in (a) to act as a sine function generator. Each ± 20 mV of input angle voltage corresponds to an input angle of $\theta = \pm 1^\circ$. Output V_o equals $1 \times \sin \theta$. (a) The AD639A is pin-programmed to output the sine of the angle voltage; (b) output voltage V_o equals the sine of θ if θ is represented by an angle voltage of 20 mV per angular degree.

16 and obtain a gain of +10. Then $V_o = 10 \sin \theta$. Pin 6 is a precision 1.80-V reference voltage that corresponds to an angle voltage of 90° (see Eq. (6-13)). We analyze sine function operation by an example.

Example 6-12

Calculate the required input angle voltage and resultant output voltage for angles of (a) $\pm 45^\circ$; (b) $\pm 90^\circ$; (c) $\pm 225^\circ$; (d) $\pm 405^\circ$.

Solution From Eq. (6-13) and Fig. 6-11,

$$(a) V_{\text{ang}} = \left(\frac{20 \text{ mV}}{1^\circ} \right) (\pm 45^\circ) = \pm 0.90 \text{ V}, V_o = 1 \sin (\pm 45^\circ) = \pm 0.707 \text{ V}.$$

$$(b) V_{\text{ang}} = \left(\frac{20 \text{ mV}}{1^\circ} \right) (\pm 90^\circ) = \pm 1.80 \text{ V}, V_o = 1 \sin (\pm 90^\circ) = \pm 1.0 \text{ V}.$$

$$(c) V_{\text{ang}} = \left(\frac{20 \text{ mV}}{1^\circ} \right) (\pm 225^\circ) = \pm 4.50 \text{ V}, V_o = 1 \sin (\pm 225^\circ) = \pm 0.707 \text{ V}.$$

$$(d) V_{\text{ang}} = \left(\frac{20 \text{ mV}}{1^\circ} \right) (\pm 405^\circ) = \pm 8.10 \text{ V}, V_o = 1 \sin (\pm 405^\circ) = \pm 0.707 \text{ V}.$$

Example 6-12 clearly illustrates that the AD639, remarkable as it is, cannot output the sine of, for example, $36,000^\circ$. This would require an angle voltage of 720 V. The normal ± 15 -V supply limits the guaranteed usable input angle to $\pm 500^\circ$, or $\pm 10,000$ V. We extend the results of Example 6-12 to summarize briefly the performance of the sine function generator in Table 6-1 and Fig. 6-11(b).

In Fig. 6-11(b), V_o is plotted against both V_{ang} and θ . A study of this figure shows that if V_{ang} *could be varied linearly* by a triangle wave, V_o *would vary sinusoidally*. Further, if the frequency of the triangle wave could be varied easily, the sine-wave frequency could *easily* be tuned, adjusted, or varied. We pursue this observation in the next section.

6-9 PRECISION SINE-WAVE GENERATOR

6-9.1 Circuit Operation

Connect the precision triangle-wave oscillator in Fig. 6-10 to the sine function generator in Fig. 6-11 to construct the precision sine-wave generator in Fig. 6-12. As a bonus, we also have precision triangle-wave and square-wave outputs. The 1.80-V reference voltage of the AD639 is connected to modulation inputs 16 and 17 of the AD630 modulator (Fig. 6-9). Circuit operation is now examined by reference to Fig. 6-12.

TABLE 6-1 AD639 Sine Functions^a

Input		Output (V)	
θ (angular degrees)	Angle voltage, V_{ang} (V)	$V_o = 1 \sin \theta$ (wire pin 10 to 9)	$V_o = 10 \sin \theta$ (wire pin 10 to 16)
0	0.00	0.000	0.000
± 45	± 0.90	± 0.707	± 7.07
± 90	± 1.80	± 1.000	± 10.07
± 135	± 2.70	± 0.707	± 7.07
± 180	± 3.60	0.000	0.000
± 225	± 4.50	± 0.707	± 7.07
± 270	± 5.40	± 1.000	± 10.00
± 315	± 6.30	± 0.707	± 7.07
± 360	± 7.20	0.000	0.00
± 405	± 8.10	± 0.707	± 7.07
± 450	± 9.00	± 1.000	± 10.00
± 495	± 9.90	± 0.707	± 7.07
± 500	± 10.00	± 0.643	± 6.43

^a Connect terminal 10 to 9 to pin program $V_o = \sin \theta$; or connect pin 10 to 16 to pin program $V_o = 10 \sin \theta$. Input angle voltage $V_{\text{ang}} = (20 \text{ mV}/1^\circ\text{C}) \theta$.

Triangle-wave rise time, 0 to $T/2$ in Fig. 6-12(b)

1. AD630
 - a. Pin 13 is at $-V_{\text{ref}} = -1.8 \text{ V}$, causing
 - b. Pin 9 to select gain = -1 to hold 13 at -1.8 V and
 - c. Op amp output voltage to ramp up.
2. Op amp
 - a. Pin 6 ramps from $-V_{\text{ref}} = -1.8 \text{ V}$ toward $+V_{\text{ref}} = 1.8 \text{ V}$ to
 - b. Hold pin 9 of the AD630 negative and
 - c. Drive input 1 of the AD639 with an angle voltage linearly from -1.8 to 1.8 V .
3. AD639
 - a. Pin 1's input angle voltage corresponds to an input angle that varies linearly from -90° to $+90^\circ$.
 - b. Pin 13 outputs $V_o = 10 \sin \theta$ from -10 to $+10 \text{ V}$.

When op amp pin 6 reaches $+1.8 \text{ V}$, pin 9 of the AD630 goes positive to select a gain of $+1$. Its output, in 13, snaps to $+1.8 \text{ V}$. This begins the *fall time*.

Triangle-wave fall time, $T/2$ to T in Fig. 6-12(b)

1. AD630: Causes the triangle wave to ramp down from $+1.8 \text{ V}$ to -1.8 V . At -1.8 V , gain is switched to -1 and a new cycle begins.

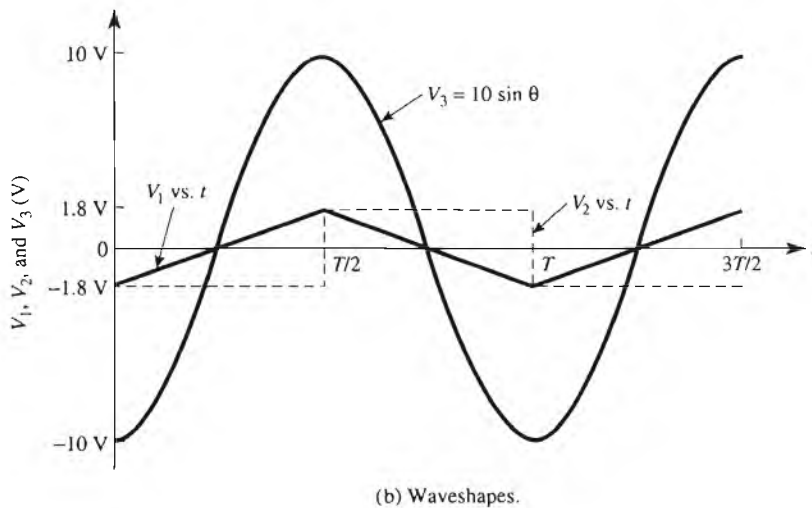
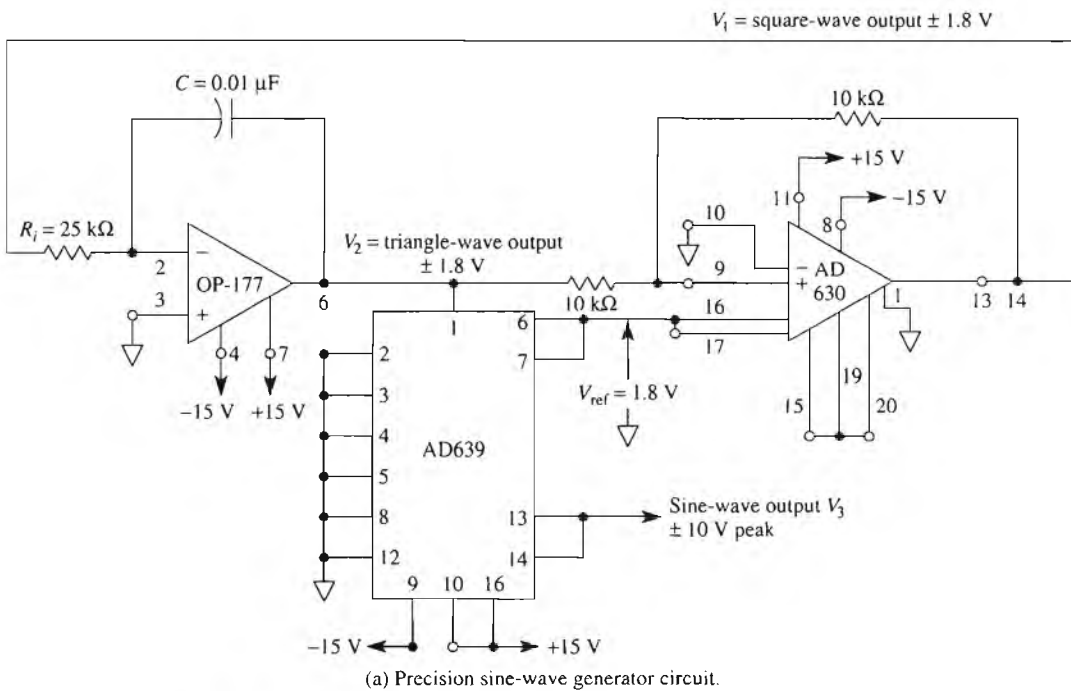


FIGURE 6-12 Frequency of the precision sine-square-triangle-wave generator in (a) can be easily changed by adjusting R_1 . Output waveshapes are shown in (b). Their amplitudes are independent of frequency.

2. *Op amp*: Applies an angle voltage to input pin 1 of the AD639 that varies linearly from +1.8 to -1.8 V.
3. *AD639*: Its input angle voltage corresponds to an input angle of $\theta = +90^\circ$ to -90° . Pin 13 outputs a sine wave that varies from +10 to -10 V.

6-9.2 Frequency of Oscillation

The frequency of oscillation, f , is determined by R_i , C , and the op amp in Fig. 6-12(a) from

$$f = \frac{1}{4R_i C} \quad (6-14)$$

Peak amplitudes of the triangle wave and square wave are precisely equal to ± 1.8 V. The sine wave has peak amplitude of ± 10 V and is synchronized to the triangle wave (for the ± 1 -V peak, change the AD639 pin 10 connection to $-V_s$).

Example 6-13

Let $C = 0.025 \mu\text{F}$ in Fig. 6-12(a) (two $0.05\text{-}\mu\text{F}$ capacitors in series). How does frequency change as R_i is changed from $10 \text{ k}\Omega$ to $100 \text{ k}\Omega$?

Solution From Eq. (6-14),

$$f = \frac{1}{4(10 \text{ k}\Omega)(0.025 \mu\text{F})} = 1 \text{ kHz} \quad f = \frac{1}{4(100 \text{ k}\Omega)(0.025 \mu\text{F})} = 100 \text{ Hz}$$

Example 6-13 shows the overwhelming superiority of this multiwave generator. Frequency is tuned easily *and* with precision. Although we have used the AD639 to generate a sine wave, this chip is a universal trigonometric function generator and could be rewired to produce other trigonometric waveforms.

6-9.3 High-Frequency Waveform Generator

Maxim's MAC038 is a 20-pin, high-frequency, precision function generator whose frequency can be controlled over a wide range from 0.1 Hz to 20 MHz. It can produce sine, triangle, or square waves at outputs that are selected by an appropriate code at two (transistor-transistor logic) TTL-compatible select inputs. It can also produce sawtooth or pulse outputs as well as a synchronizing output.

Applications for this versatile chip include function or FSK generation, VCOs (voltage controlled oscillator), frequency modulators, and synthesizers, as well as pulse-width modulation.

6-10 PSpice SIMULATION OF SIGNAL GENERATOR CIRCUITS

In this section, we will use PSpice to model and simulate the performance of four signal generator circuits studied in this chapter: the free-running multivibrator, the one-shot multivibrator, the bipolar triangle-wave generator, and the unipolar triangle-wave generator.

6-10.1 Free-Running Multivibrator

Refer to Fig. 6-1 and create the PSpice model of the circuit. Set the resistor and capacitor values as given in Examples 6-1 and 6-2. Use the 741 op amp if you are using the evaluation software package of PSpice. Obtain a plot of V_C and V_o versus time. For the circuit to begin oscillating in a simulation it is necessary to provide a sudden impulse at the beginning of the simulation. This stimulus can be generated by using two pulse sources instead of two dc sources to power the op amp. The pulse width will be set to a much longer time interval than the period of oscillation and will have a fast rise time to simulate suddenly applying power to the circuit. To begin, place the following parts in the work area.

Draw => Get New Part

Part	Number	Library
=> uA741	1	eval.slb
=> VPULSE	2	source.slb
=> R	3	analog.slb
=> C	1	analog.slb
=> GLOBAL	4	port.slb
=> AGND	4	port.slb

(Note: We are using **VPULSE** instead of **VDC** for the op amp supplies.) Arrange the parts and wire the circuit as shown in Fig. 6-1. Change the attributes of the parts as given in Examples 6-1 and 6-2. Set up each of the **VPULSE** attributes by double-clicking on the symbol. In the pop-up box, set the values for **V1** (minimum input voltage), **V2** (maximum input voltage), **TD** (time delay), **TR** (rise time), **TF** (fall time), **PW** (pulse width), and **PER** (period):

V1 => 0V => Save Attr

V2 => 15V => Save Attr => Change Display => Both name and value

TD => 0 => Save Attr

TR => 1ns => Save Attr

TF => 1ns => Save Attr

PW => 50s => Save Attr

PER => 51s => Save Attr

Each pulse attribute pop-up box is set up the same because the $-V$ supply has been rotated 180 degrees, as we have done with the dc supplies connected to pin 4 of other op amps. Double-click on the lead from the capacitor and label it **Vc**. Double-click on the lead from the output terminal of the op amp and label it **Vo** (see Fig. 6-13).

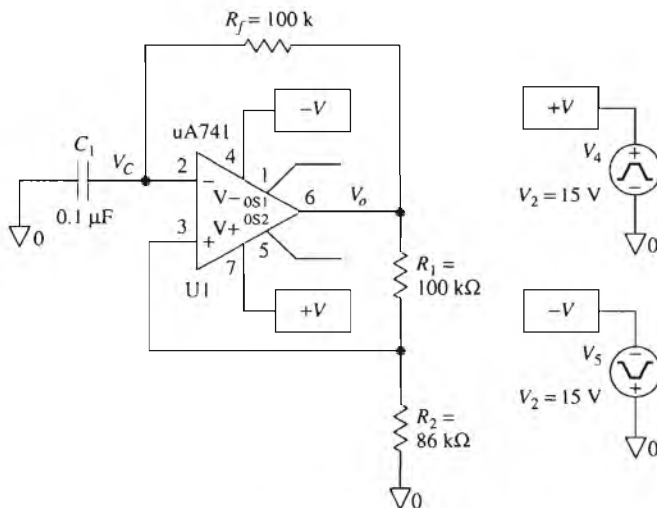


FIGURE 6-13 PSpice model of the free-running multivibrator.

In order to obtain a plot of V_C and V_o versus time, we must initialize the transient menu.

Analysis => Setup => Select Transient
Click Transient => Print Step: => 0.2ms
=> Final Time: => 60ms

Save the circuit as a file with the .SCH extension. Run the simulation

Analysis => Simulate

In the Probe window, select

Trace => Add => V[Vc]
=> V[Vo]

Label the plots and obtain a printout as shown in Fig. 6-14.

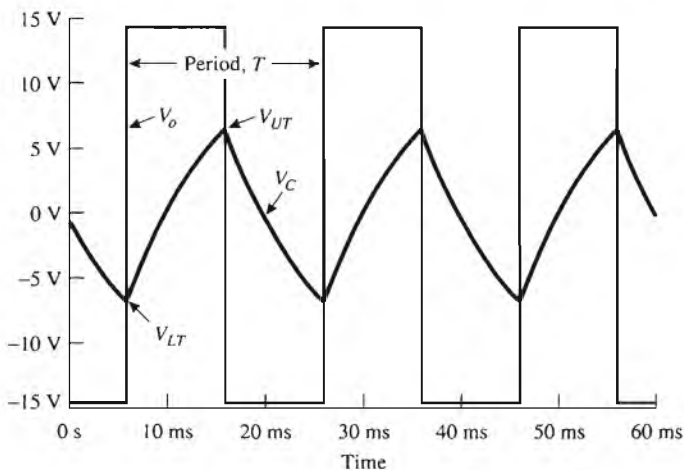


FIGURE 6-14 Capacitor and output voltage waveforms generated from the PSpice model of Fig. 6-13.

6-10.2 One-Shot Multivibrator

Create the PSpice model of the one-shot multivibrator shown in Fig. 6-4. Use the 741 op amp if you are using the evaluation software package of PSpice. Obtain a plot of E_i and V_o versus time. Place the following parts in the work area.

Draw => Get New Part

Part	Number	Library
=> uA741	1	eval.slb
=> D1N4002	2	eval.slb
=> VPULSE	1	source.slb
=> VDC	2	source.slb
=> R	4	analog.slb
=> C	2	analog.slb
=> GLOBAL	4	port.slb
=> AGND	6	port.slb

Arrange the parts and wire the circuit as shown in Fig. 6-4. Change the attributes as given in Fig. 6-4, but set $R_2 = 2 \text{ k}\Omega$ [so that results closely approximate Eq. (6-4a)] and set **VPULSE** for a 10-V negative pulse with a pulse width of 0.1 ms. The pulse attributes are shown in Fig. 6-15. Double-click the **VPULSE** symbol and set the values accordingly. Double-click on the lead from **VPULSE** and label it E_i . Label the output of the op amp as V_o . See Fig. 6-15.

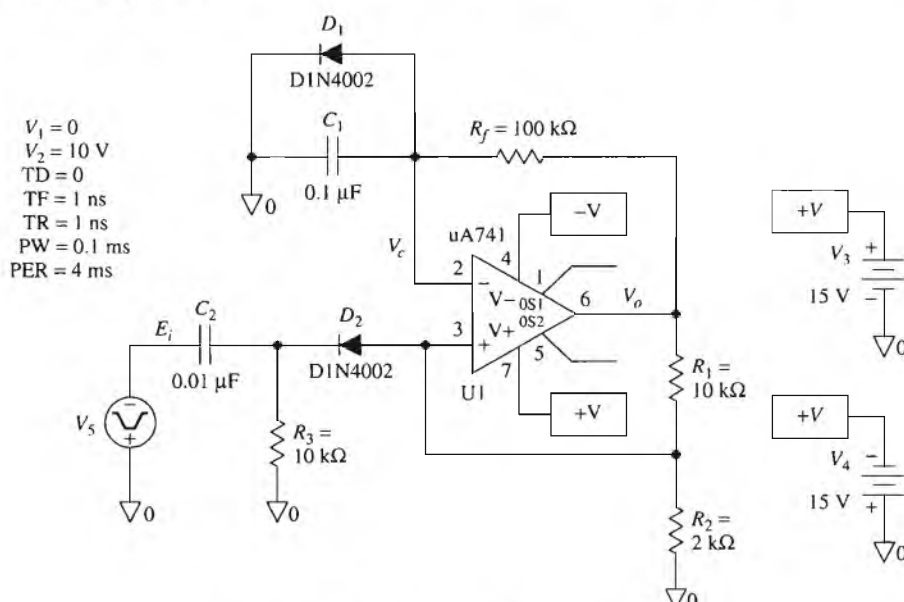


FIGURE 6-15 PSpice model of a one-shot multivibrator.

In order to obtain a plot of E_i and V_o versus time, we must initialize the transient menu.

Analysis => Setup => Select Transient
Click **Transient** => **Print Step: => 100us**
=> **Final Time: => 10ms**

Save the circuit as a file with the .SCH extension. Run the simulation

Analysis => Simulate

In the Probe window, select

Trace => Add => V[Ei]
=> **V[Vo]**

Label the plots and obtain a printout as shown in Fig. 6-16.

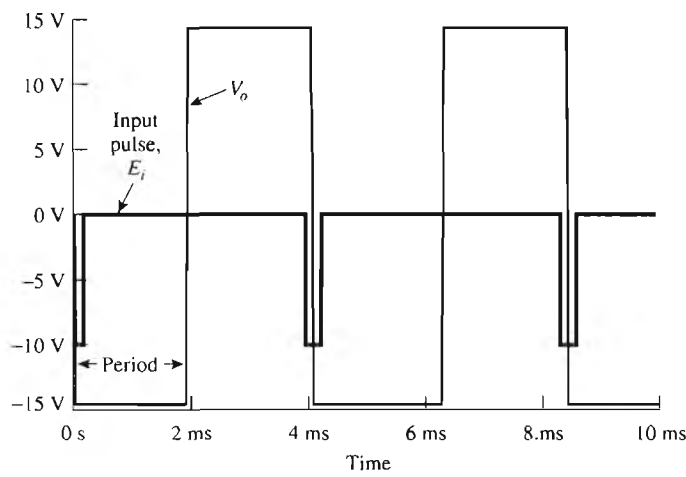


FIGURE 6-16 Waveforms for the one-shot multivibrator modeled in PSpice in Fig. 6-15.

6-10.3 Bipolar Triangle-Wave Generator

Create the PSpice model of the bipolar triangle-wave generator shown in Fig. 6-6. Use two 741 op amps. Obtain a plot of V_A and V_o versus time. Place the following parts in the work area.

Draw => Get New Part

Part	Number	Library
=> uA741	2	eval.slb
=> VPULSE	2	source.slb
=> R	4	analog.slb
=> C	1	analog.slb
=> GLOBAL	6	port.slb
=> AGND	5	port.slb

Arrange the parts, change the attributes, and wire the circuit as shown in Fig. 6-6. The **VPULSE** attributes are set the same as they are in Section 6-10.1 (see Fig. 6-17).

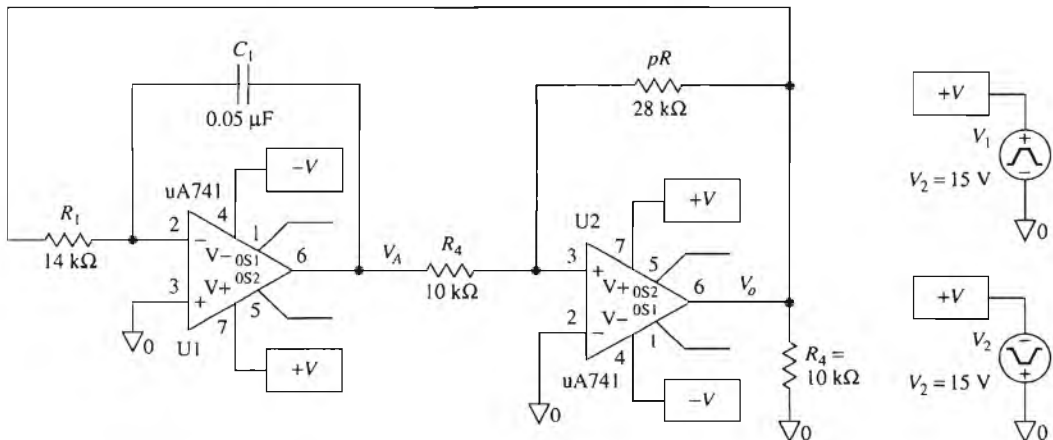


FIGURE 6-17 Bipolar triangle-wave generator modeled in PSpice.

In order to obtain a plot of V_A and V_o versus time, set the transient menu.

Analysis => Setup => Select Transient

Click **Transient** => **Print Step:** => **0.01ms**

=> **Final Time:** => **3ms**

Save the circuit as a file with the **.SCH** extension. Run the simulation

Analysis = > Simulate

In the Probe window, select

Trace => Add => V[VA]

=> **V[Vo]**

Label the plots and obtain a printout as shown in Fig. 6-18.

6-10.4 Unipolar Triangle-Wave Generator

Modify the PSpice model of Fig. 6-17 to create the unipolar triangle-wave generator shown in Fig. 6-7. From the PSpice parts list, obtain a 1N4002 diode and place it in series with pR (see Fig. 6-19). Save the circuit with the **.SCH** extension.

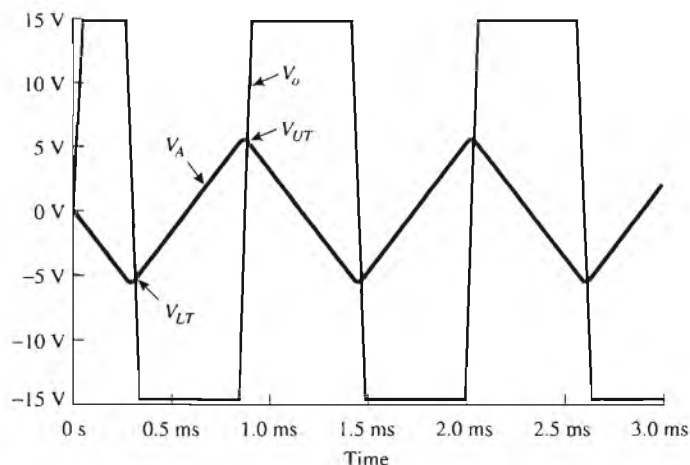


FIGURE 6-18 Waveforms generated from the bipolar triangle-wave generator shown in Fig. 6-17.

Follow the procedure given in Section 6-10.3 to obtain a plot of V_A and V_o (see Fig. 6-20).

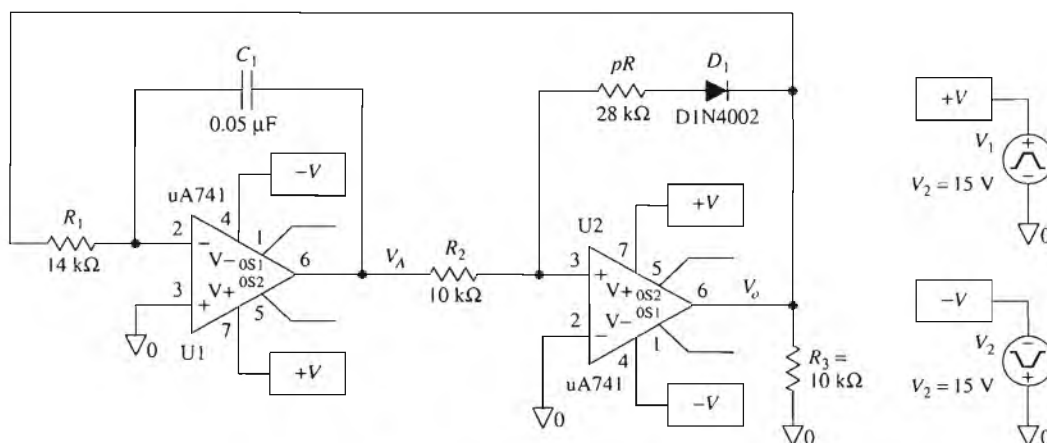


FIGURE 6-19 Unipolar triangle-wave generator modeled in PSpice.

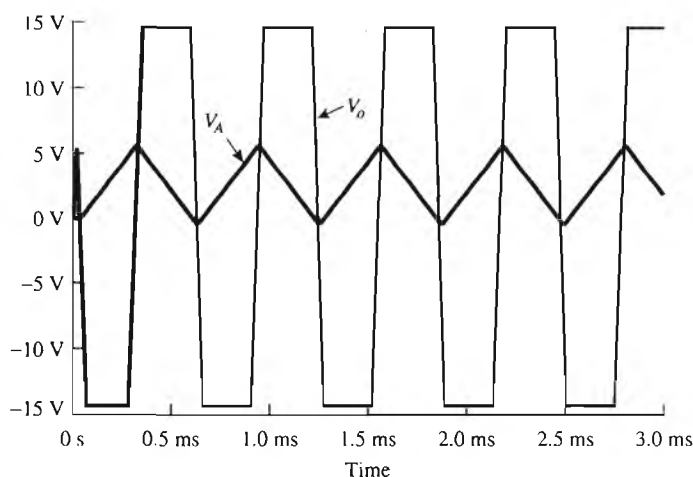


FIGURE 6-20 Waveforms V_A and V_o for the unipolar triangle-wave generator of Fig. 6-19.

PROBLEMS

- 6-1. Make two drawings of a multivibrator circuit with $R_1 = 100 \text{ k}\Omega$, $R_2 = 86 \text{ k}\Omega$, $R_f = 10 \text{ k}\Omega$, and $C = 0.01 \text{ }\mu\text{F}$. Show the direction of current through C and calculate both V_{UT} and V_{LT} for (a) $V_o = +V_{\text{sat}} = 15 \text{ V}$; (b) $V_o = -V_{\text{sat}} = -15 \text{ V}$.
- 6-2. Calculate the frequency of oscillation for the multivibrator circuit in Problem 6-1.
- 6-3. In Problem 6-1, if C is changed to $0.1 \text{ }\mu\text{F}$, do you expect the output frequency to oscillate at 500 Hz? (See Example 6-3.) What could you do to R_f to increase frequency to 1000 Hz?
- 6-4. The monostable multivibrator of Figs. 6-4 and 6-5 generates a negative output pulse in response to a negative-going input signal. How would you change these circuits to get a positive output pulse for a positive-going input edge?
- 6-5. Explain what is meant by *monostable recovery time*.
- 6-6. Sketch a one-shot multivibrator circuit whose output will deliver a negative pulse lasting 1 ms with a recovery time of about 0.1 ms.
- 6-7. Assume for simplicity that saturation voltages in the triangle-wave oscillator of Fig. 6-6 are $\pm 15 \text{ V}$, $R_i = R = 10 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$, and $pR = 50 \text{ k}\Omega$. Find the peak triangle-wave voltages and oscillating frequency.
- 6-8. Refer to the triangular-wave oscillator circuit of Fig. 6-6. What happens to peak output voltages and oscillating frequency if you (a) double pR only; (b) double R_i only; (c) double capacitor C only?
- 6-9. Change pR to $14 \text{ k}\Omega$ and C to $0.1 \text{ }\mu\text{F}$ in the unipolar triangle-wave generator of Fig. 6-7. Find the resulting peak output voltage and frequency of oscillation. (See Example 6-7.)

- 6-10. In the sawtooth-wave generator of Fig. 6-8(a), let $V_{\text{ref}} = 1 \text{ V}$, $R_i = 10 \text{ k}\Omega$, and $C = 0.1 \text{ }\mu\text{F}$.
- (a) Find an expression for frequency f in terms of E_i .
 - (b) Calculate f for $E_i = 1 \text{ V}$ and $E_i = 2 \text{ V}$.
- 6-11. These questions refer to the AD630 balanced modulator circuit in Fig. 6-9.
- (a) Name the application for which the AD630 is wired.
 - (b) When pin 9 is at a positive voltage, which amplifier is selected, and what is the value of V_o ?
 - (c) Suppose that V_{ref} is a $\pm 1\text{-V}$ -peak sine wave and pin 9 is at 1 V ; what happens at V_o when pin 9 is changed to -1 V ?
- 6-12. Figure 6-10 shows a precision triangle/square-wave oscillator. Three components control peak output voltages and oscillating frequency, R_p , C , and V_{ref} .
- (a) Which does what?
 - (b) Can the oscillating frequency be adjusted independent of peak outputs, and vice versa?
 - (c) What must be done to change the frequency from 100 to 500 Hz and the peak voltages from $\pm 5 \text{ V}$ to $\pm 1 \text{ V}$?
- 6-13. $V_o = 0.866 \text{ V}$ in the sine function generator circuit of Fig. 6-11.
- (a) What angle does this represent?
 - (b) What is the value of the input angle voltage?
- 6-14. Calculate V_o in Fig. 6-11 when the input angle is 30° and pin 10 is wired to (a) pin 9; (b) pin 16.
- 6-15. Design a sine-wave oscillator whose frequency can be varied from 0.5 Hz to 50 Hz with just a single variable resistor.